

# Rajit Sinha

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## EDUCATION

### Lebanon Trail High School

*Frisco ISD — Frisco, TX*

Expected May 2027

*GPA: 3.807/4.0 (4.937 weighted, 6.0 scale)*

- SAT: 1510 (790 Math, 720 Reading & Writing); 12 AP courses; Top 20% of a class of 551

## EXPERIENCE & LEADERSHIP

### Texas High School Aerospace Scholars (NASA) — Design Engineer

2025 – 2026

*NASA Johnson Space Center*

*Houston, TX*

- Selected for the residential Base Camp experience as 1 of 62 students statewide from 1,014 applicants, advancing through a five-month online aerospace course
- Served as Design Engineer on a nine-member team for Moonshot, an Artemis-themed design challenge, producing a Preliminary Design Review for a lunar EVA helmet and glove system balancing hazard protection against astronaut mobility alongside a NASA consultant
- Modeled the EVA glove assembly in Fusion 360 (coupler, glove entrance, finger covers) and researched its five-layer construction: bladder, restraint, coil, thermal, TMG
- Presented technical findings to NASA engineers and scientists in a poster session; completed design challenges and facility tours at JSC

### American Rocketry Challenge — Team Captain

2024 – 2026

*GEM Academy & Center*

*Plano, TX*

- Led a team through the design, build, and flight testing of a competition rocket after a year as a member
- Built the rocket's flight computer — Arduino Nano R4 and DPS310 barometric sensor wired to a servo-triggered parachute deployment
- Wrote flight software in C++ sampling at 50 Hz that low-pass filters altitude and velocity and **predicts apogee in real time** from a quadratic-drag model corrected for servo delay, deploying early to target 750 ft instead of reacting after the peak
- Added three independent deployment triggers (five-sample prediction agreement, sustained descent, altitude failsafe) and EEPROM logging so peak altitude survives power loss
- Flight tested configurations from 470–509 g on Estes F50 and AeroTech F30/F32 motors; designed recovery hardware including a custom hexagonal parachute and 3D-printed components
- Diagnosed an in-flight deployment failure as insufficient servo torque after confirming the avionics had functioned correctly

### Young Tech Pioneers — Vice President of Fundraising

2024 – Present

*Youth entrepreneurship and technology nonprofit*

*Frisco, TX*

- Took over fundraising the summer after joining as a tenth-grade member, raising money for a nonprofit teaching business and technology to K–8 students
- Helped run 10 Children's Business Fairs where 200+ kids started their own businesses and earned over \$10,000 collectively, handling vendors, volunteers, and event logistics for crowds of 150–200

## TECHNICAL PROJECTS

### Independent Technical Work | *Fusion 360, 3D Printing, Google Apps Script*

2024 – Present

- Design and print functional engineering parts: rocket hardware, structural components, and prototypes
- Built an automated scholarship tracker that sorts entries dynamically and prioritizes them by deadline

## HONORS & AWARDS

**BPA State:** 1st Digital Marketing, 2nd Digital Communication & Design, 3rd Information Tech, 5th Computer Programming (2026); State qualifier 2025

**BPA Regionals:** 3rd Computer Science Concepts, 7th Information Technology Concepts (2024)

**AP Scholar with Distinction** (2026); **AP Scholar with Honor** (2025); **UIL Solo & Ensemble State Qualifier** (2024)

**Community Service:** GEM Academy & Center volunteer, 100+ hours

## TECHNICAL SKILLS

**Languages & Tools:** Python, Java, C++, Arduino IDE, Fusion 360 (CAD), 3D printing

**Areas:** Embedded systems, real-time control logic, sensor filtering, physics modeling, failure analysis